Year 7
Mathematics

Unit 1: Analysing and Displaying Data

1.4 More Graphs and Tables 1

More Graphs and Tables 1

STARTER · 5 MIN

Date: _____ **Title: More Graphs and Tables 1**

LEARNING OBJECTIVES

- To interpret and draw line graphs

STARTER *Answer in your book. No calculators.*

1. What number is halfway between 10 and 20?
2. One small square is 2 °C. What is 3 squares above 10 °C?
3. Work out 15 − 7
4. It goes from 8 °C to 13 °C. How much warmer is it?

Starter — Answers

STARTER

Tick or correct in green pen

1. **15**
2. **16 °C**
3. **8**
4. **5 °C**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can read values accurately from a line graph	☐
SILVER	I can plot points and draw a line graph from a table of data	☐
GOLD	I can interpret what a line graph shows about the data	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Graphs and Tables 1

INTRODUCTION · 3 MIN

A line graph shows how something changes, usually over time.

Line graph

Points are joined with straight lines.

Time goes on the horizontal axis

Reading values

Go up from the axis to the line, then across.

Check what one square is worth

Plotting

Choose a scale, label both axes, plot with ×.

Join the points with a ruler

Trends

Up = increase. Down = decrease.

Steeper line = faster change

Examples 1 and 2 — Reading Line Graphs

EXAMPLES · 5 MIN

BRONZE

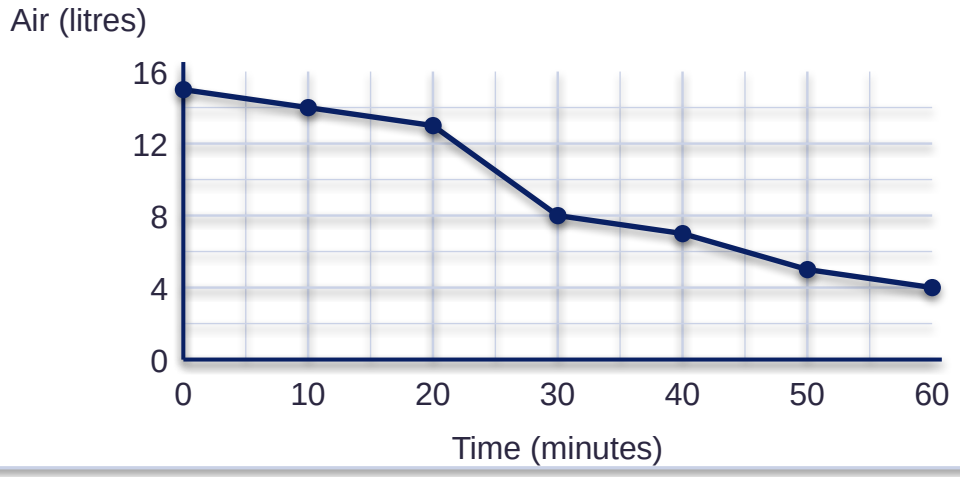
Watch and copy into your book

Hint
Up from the time, then across
Check what one square is worth

Example 1

Air in a scuba tank:

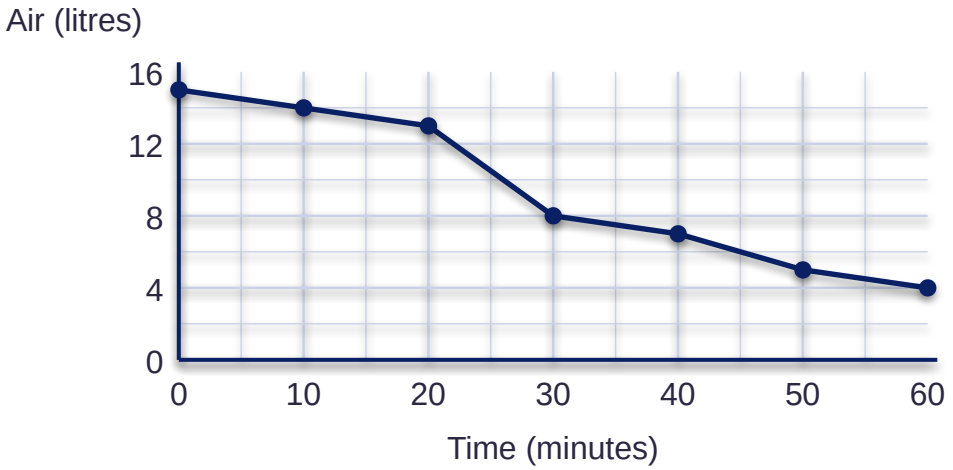
How much air was there at the start and after 20 minutes?



Example 2

Air in a scuba tank:

When did the air drop to 7 litres?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

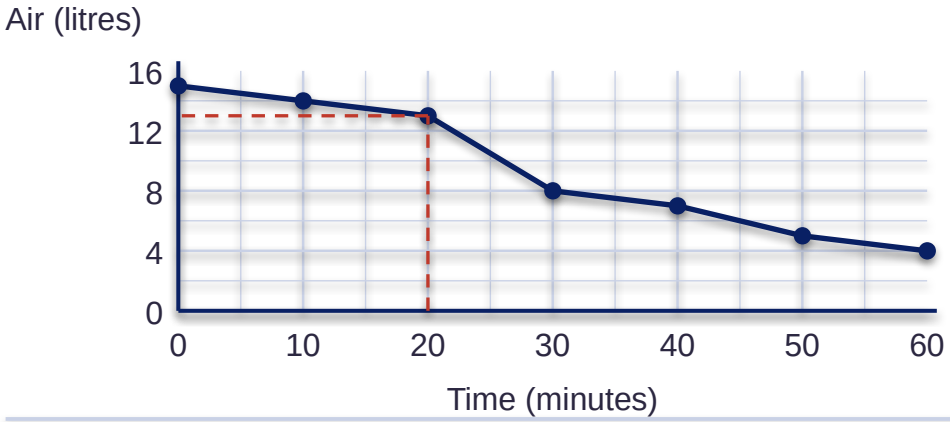
Check your copy and correct in green pen

Example 1

Up from the time, then across

At the start = 15 litres

After 20 minutes = 13 litres

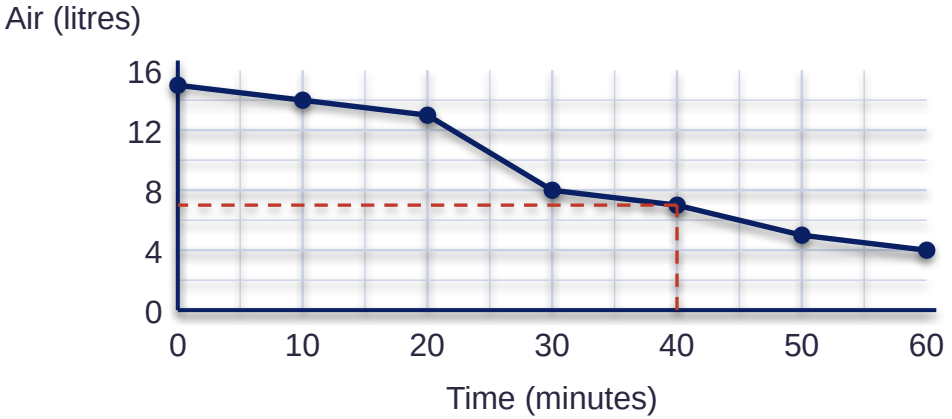


Tip: one small square is 2 litres.

Example 2

Across from 7 litres, then down

Time = 40 minutes



Tip: one small square is 5 minutes.

Task 1

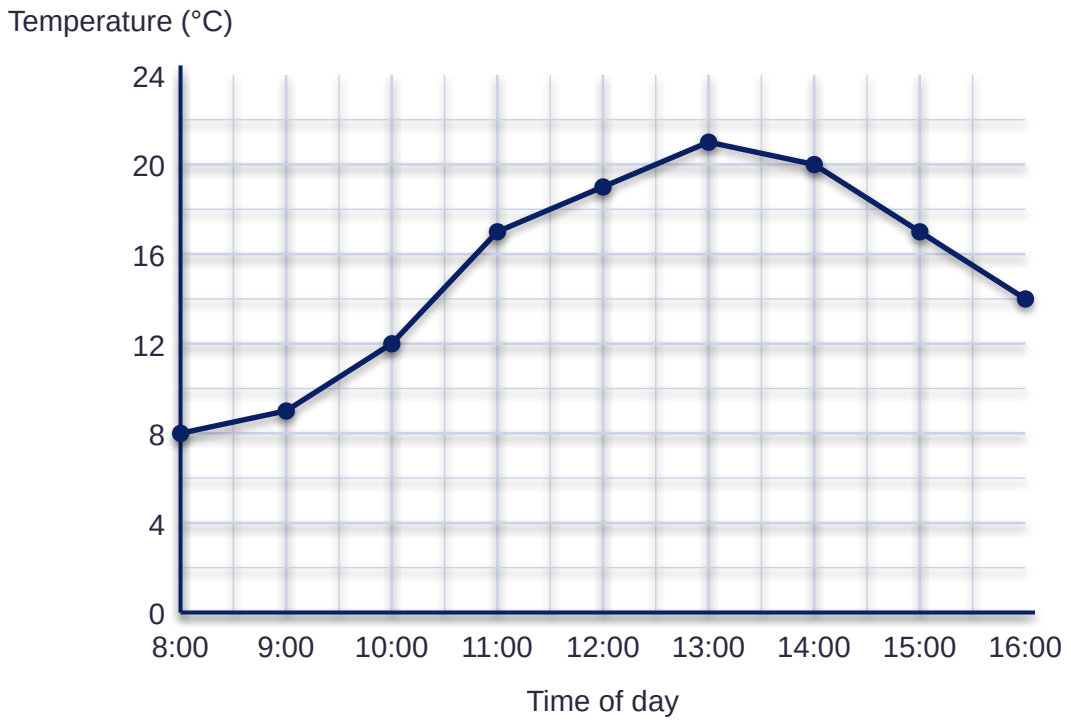
PRACTICE · 9 MIN

BRONZE

Use the line graph to answer each question

Hint
Up from the time, then across
Check what one square is worth

Temperature in a garden



1. What was the temperature at 10:00?
2. What was the highest temperature?
3. When was it first 17 °C?
4. What was the temperature at 16:00?
5. How much did it rise from 8:00 to 12:00?

★ In which hour did it rise the most?

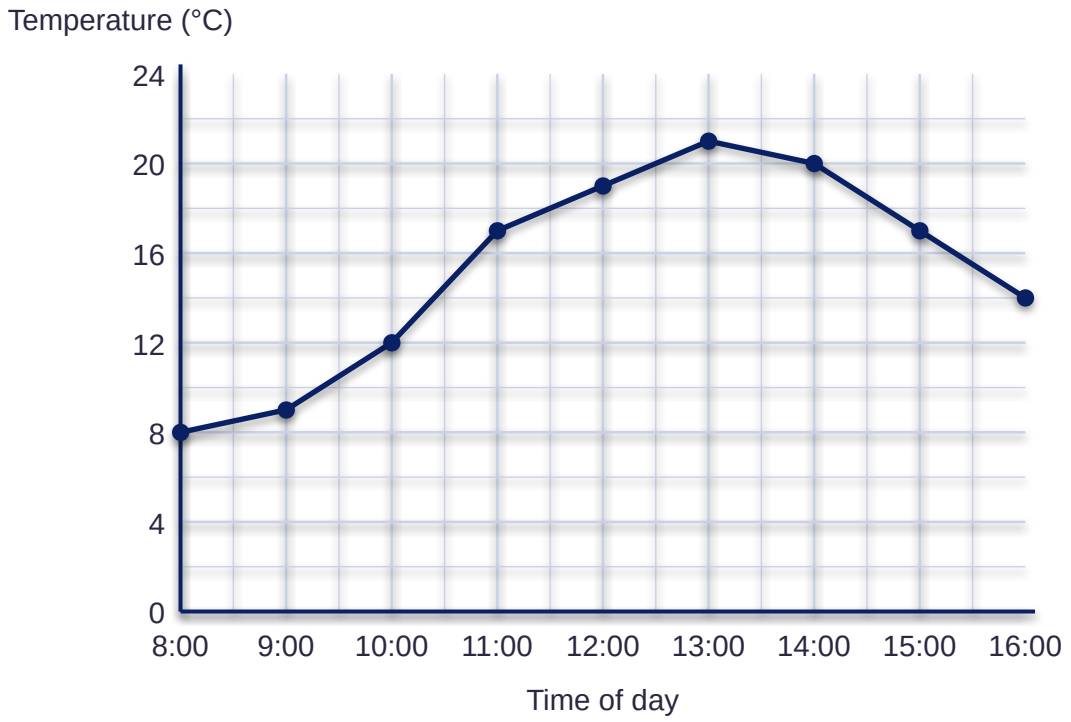
Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

Temperature in a garden



1. **12 °C**
2. **21 °C (at 13:00)**
3. **11:00**
4. **14 °C**
5. **Rise = 19 – 8 = 11 °C**

★ **10:00 to 11:00 (5 °C)**

Examples 3 and 4 — Drawing Line Graphs

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Time on the horizontal axis Plot with ×, join with a ruler

Example 3

Wind speed each day (knots):

Draw a line graph.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Speed	12	18	25	20	9	14	22

Example 4

Average maximum (°C):

Draw both lines on one graph.

Month	Jan	Feb	Mar	Apr	May	Jun
Canberra	28	27	24	20	15	12
London	8	9	11	14	18	21

Examples 3 and 4 — Corrections

CORRECTIONS

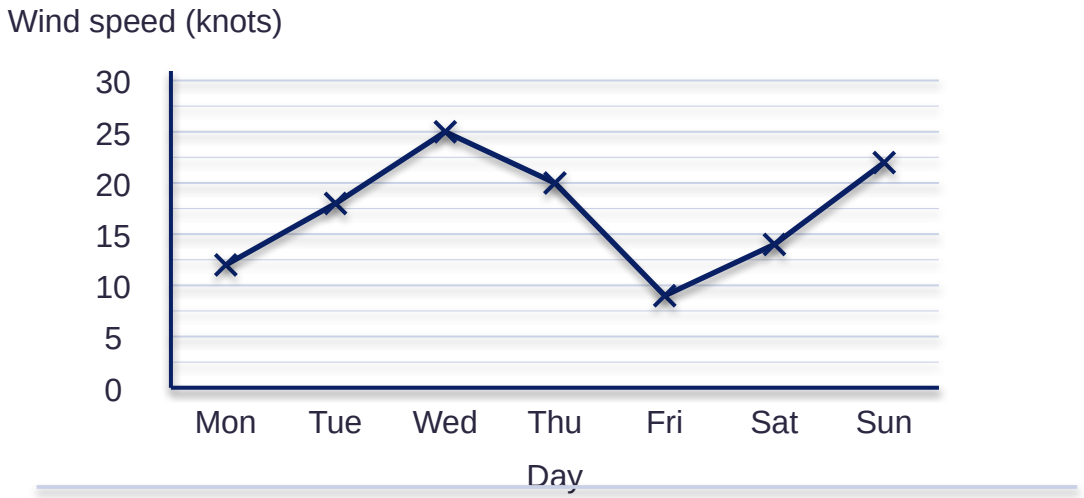
SILVER

Check your copy and correct in green pen

Example 3

Days across, speed up

One line = 5 knots

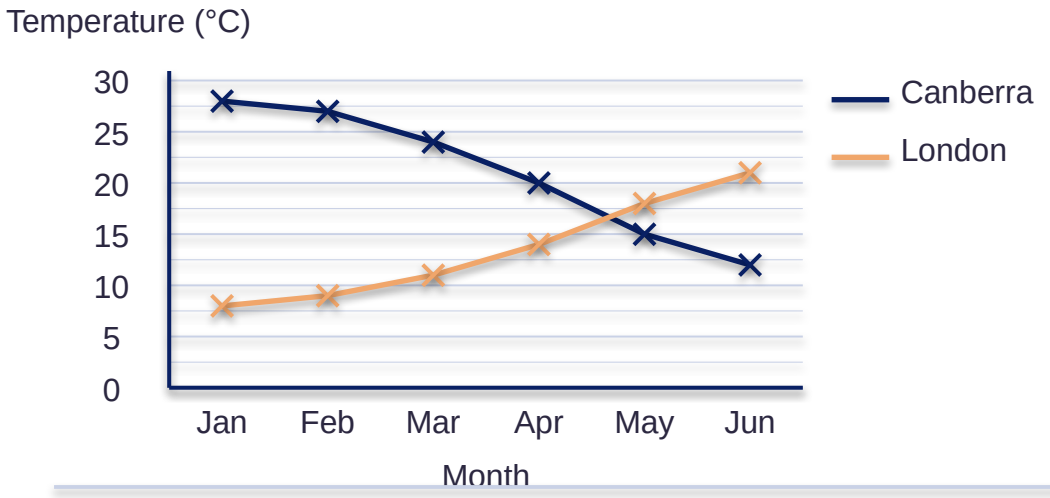


Tip: plot with ×, join with straight lines.

Example 4

Two lines need a key

One line = 5 °C



Tip: label both axes and both lines.

Task 2

PRACTICE · 9 MIN

SILVER

Draw a line graph of the table in your book

Hint Time on the horizontal axis
Plot with ×, join with a ruler

Height of a sunflower

Week	Height (cm)
0	10
1	18
2	30
3	45
4	62
5	75
6	82

1. Which goes on the horizontal axis?
2. Choose a scale for the height axis.
3. Plot the points and join them.
4. Estimate the height after $2\frac{1}{2}$ weeks.
5. In which week did it grow the most?

★ Will it keep growing this fast? Explain.

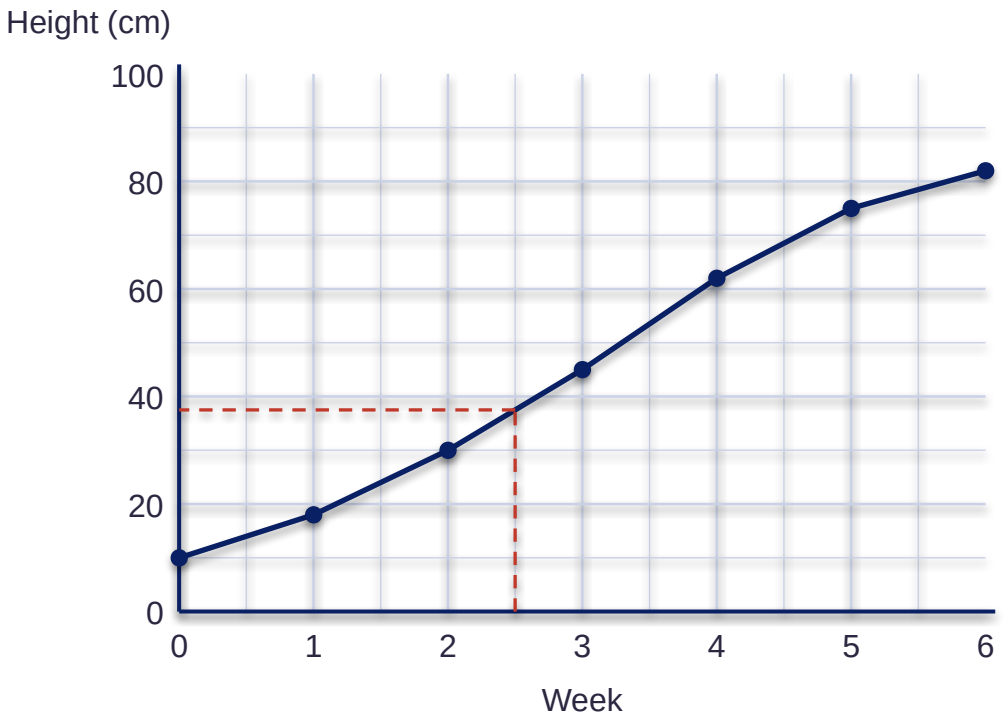
Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

Height of a sunflower



1. **Week (time)**
2. **0 to 100, one line = 20 cm**
3. **See the graph**
4. **About 38 cm**
5. **Week 3 to 4: 62 – 45 = 17 cm**

★ **No: it grew 13 cm, then only 7 cm**

Examples 5 and 6 — Interpreting Line Graphs

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Steeper = faster change Compare: give both values

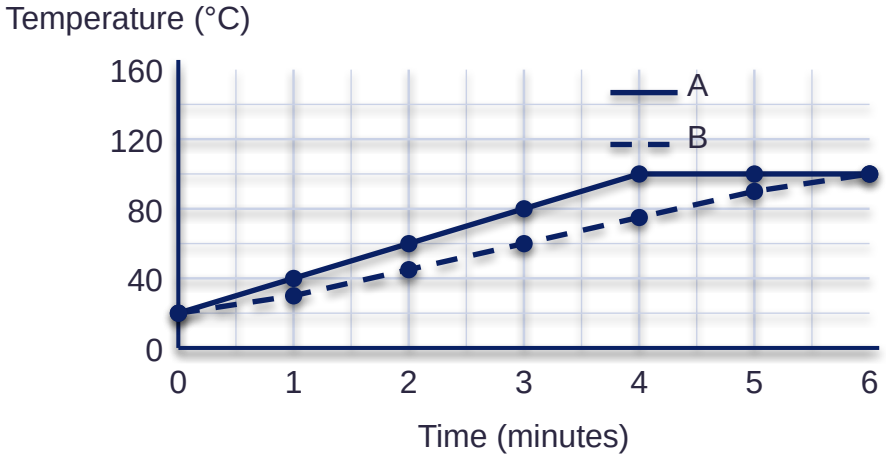
Example 5

(2 marks)

Two liquids are heated.

Write down the room temperature.

Find the difference at 3 minutes.

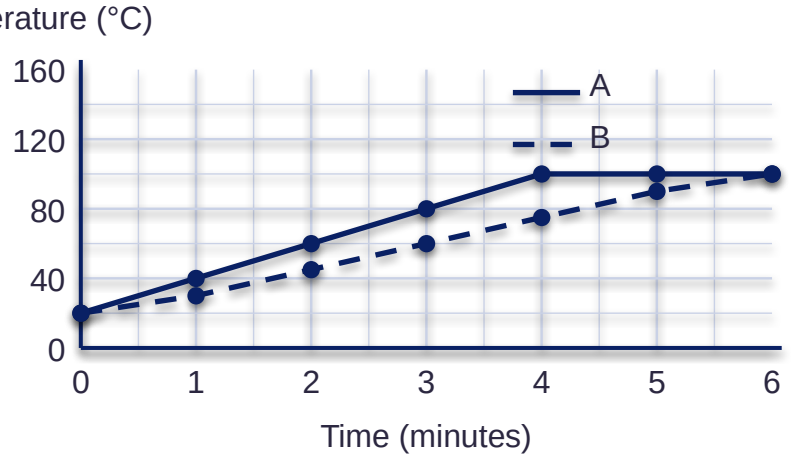


Example 6

(2 marks)

Use the same graph.

Compare the times taken to reach 100 °C.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Read where the lines start

Room temperature = 20 °C

Difference = 80 − 60

= 20 °C

Tip: at 0 minutes both are at room temperature.

Example 6

Read where each line reaches 100

A = 4 minutes

B = 6 minutes

A took 2 minutes less than B.

Tip: give both times, then compare.

Task 3

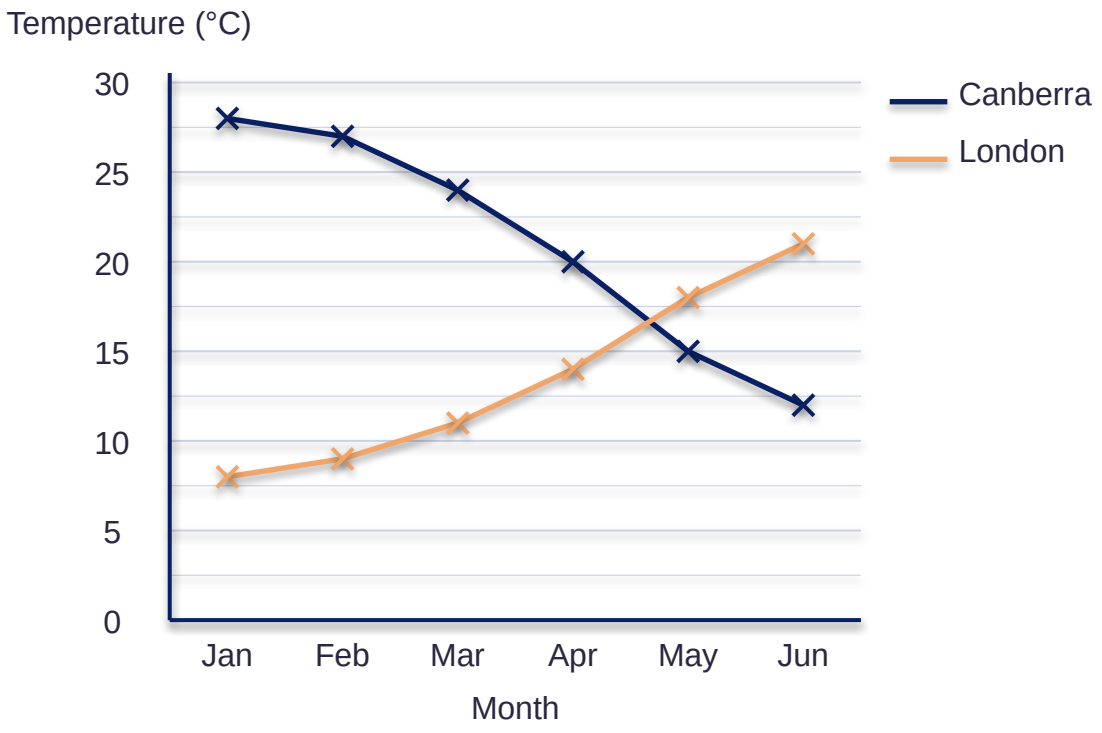
PRACTICE · 9 MIN

GOLD

Use the line graph of temperatures

Hint Steeper = faster change Compare: give both values

Average maximum temperature



1. What was the highest temperature in Canberra?
2. Which month was coldest in London?
3. Between which months do the lines cross?
4. Find the difference in January.
5. Work out the range for London.

★ Write a sentence comparing the two cities.

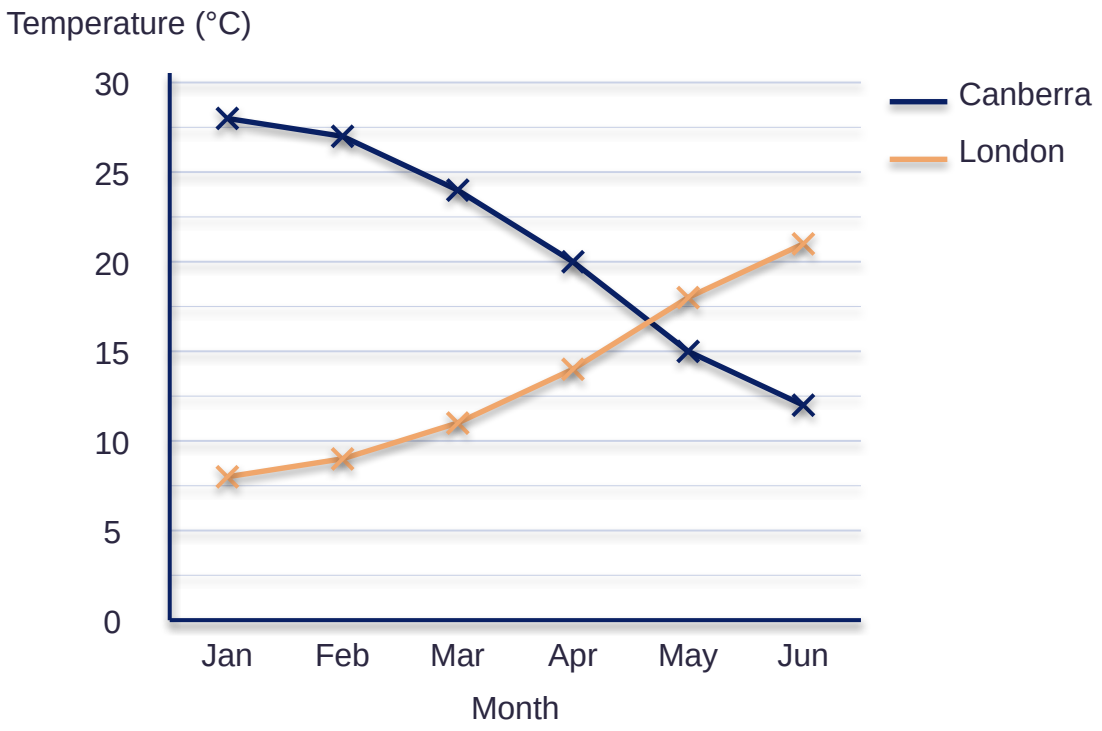
Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

Average maximum temperature



1. 28 °C (January)
2. January (8 °C)
3. April and May
4. Difference = 28 − 8 = 20 °C
5. Range = 21 − 8 = 13 °C

★ Canberra gets colder while London gets warmer.

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE On a line graph, which axis usually shows time?

SILVER 10 °C at 2 pm, 16 °C at 3 pm. How much did it rise?

GOLD One line is steeper than another. What does that tell you?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **The horizontal axis**

SILVER **6 °C**

GOLD **It is changing faster**

Exam Practice — Reading Line Graphs

EXAM PRACTICE · 10 MIN

BRONZE

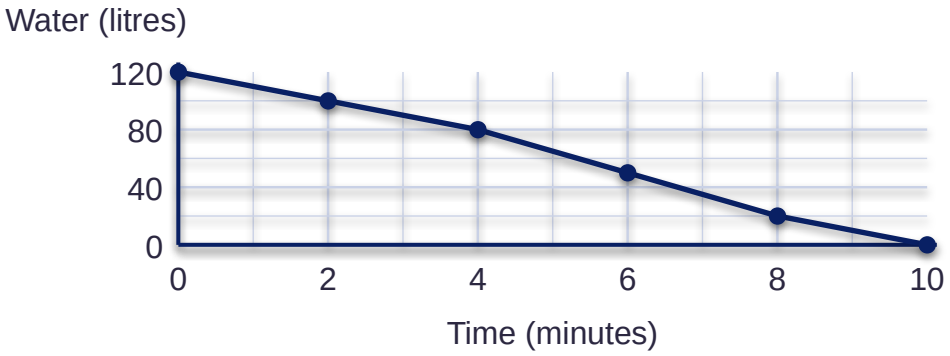
IGCSE exam-style: show all your working

Hint
Up from the time, then across
Check what one square is worth

Question 1

(2 marks)

Water in a bath as it empties.



(a) How much water at the start? (1)

..... litres

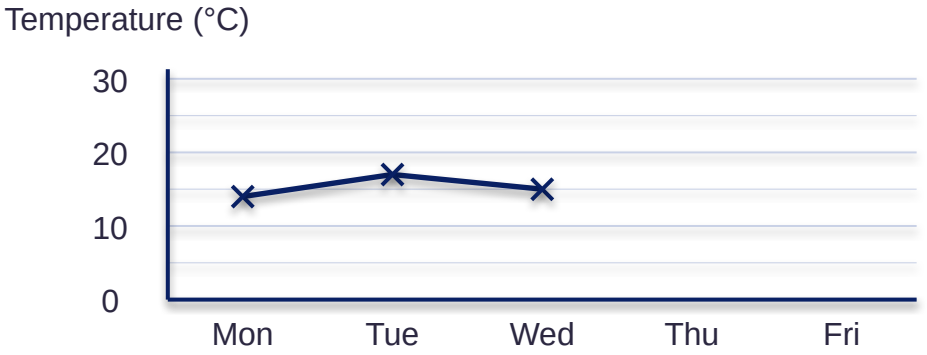
(b) How much was left at 6 minutes? (1)

..... litres

Question 2

(3 marks)

Temperatures at noon, in °C.



(a) Plot Thu 19 °C and Fri 22 °C. (2)

(b) Which day was the warmest? (1)

.....

Exam Practice — Drawing Line Graphs

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

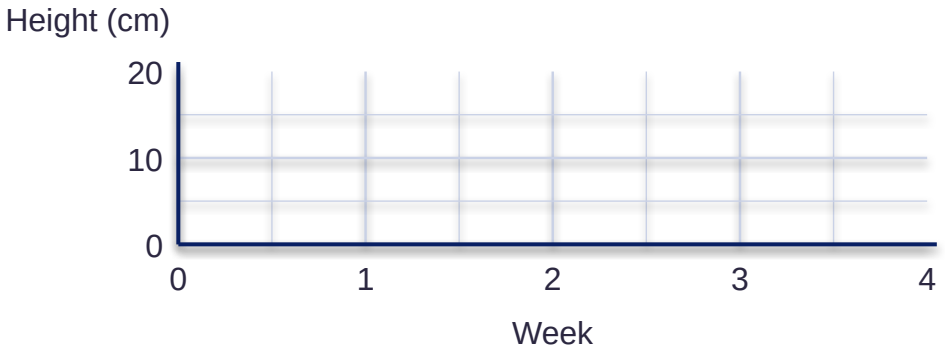
Hint Time on the horizontal axis
Plot with ×, join with a ruler

Question 3

(3 marks)

Height of a plant (cm):

Week 0: 4, 1: 7, 2: 12, 3: 14, 4: 15



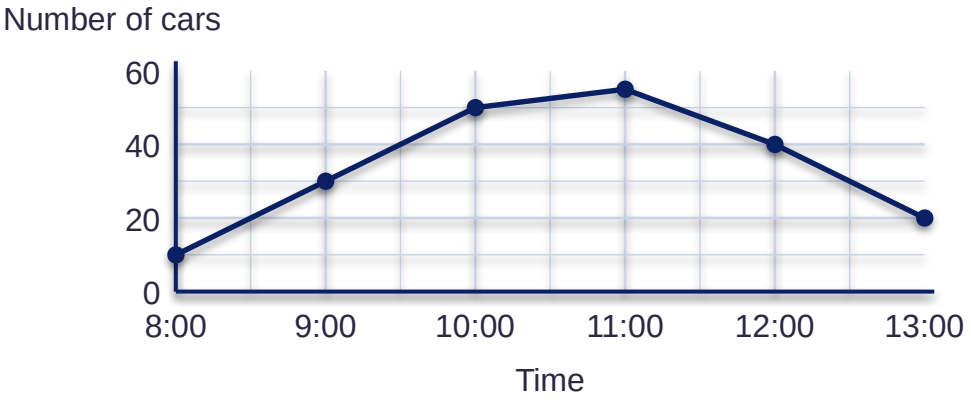
(a) Draw the line graph. (2)

(b) In which week did it grow most? (1)

Question 4

(2 marks)

Cars in a car park.



(a) How many cars at 9:00? (1)

.....

(b) When were there most cars? (1)

.....

Exam Practice — Interpreting Line Graphs

EXAM PRACTICE · 10 MIN

GOLD

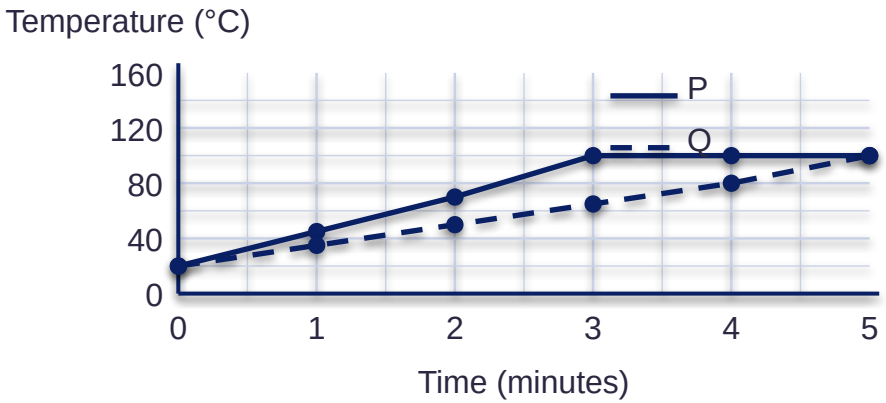
IGCSE exam-style: show all your working

Hint
Steeper = faster change
Compare: give both values

Question 5

(3 marks)

Two liquids are heated.



(a) Write down room temperature. (1)

..... °C

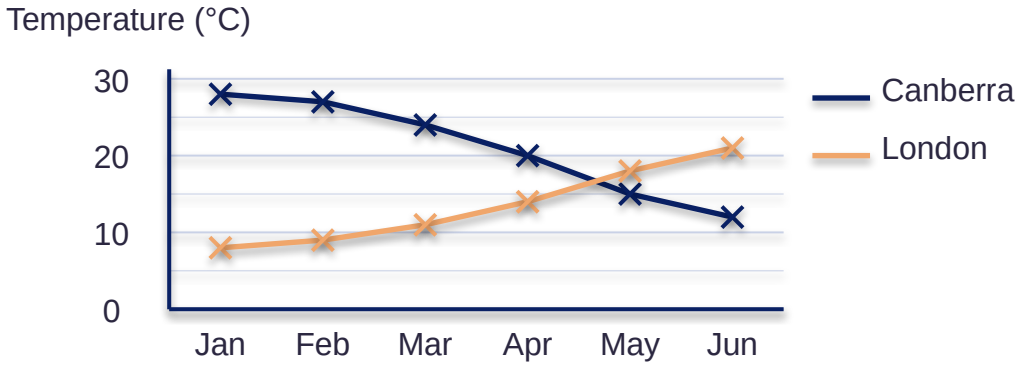
(b) Compare times to reach 100 °C. (2)

.....

Question 6

(2 marks)

Average maximum temperature.



(a) Find the difference in March. (1)

..... °C

(b) Warmest month in London? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 120 litres B1

(b) 50 litres B1

Tip: one small square is 20 litres.

Question 2

(a) × at Thu 19 and Fri 22 B1

joined with straight lines B1

(b) Friday B1

Tip: one small line is 5 °C.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 5 points plotted B1

joined with straight lines B1

(b) Week 1 to 2 (5 cm) B1

Tip: one small square is 5 cm.

Question 4

(a) 30 B1

(b) 11:00 (55 cars) B1

Tip: one small square is 10 cars.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 20 °C B1

(b) P: 3 minutes, Q: 5 minutes B1

P is 2 minutes quicker B1

Tip: give both times, then compare.

Question 6

(a) Difference = 24 − 11

= 13 °C B1

(b) June B1

Tip: read each line against the scale.